



Operating Instructions

Gas engine
12V4000L64
16V4000L64
12V4000L64FNER
16V4000L64FNER

MS15027/09E

Engine model	kW/cyl.	Application group
12V4000L64	130 kW/cyl.	3A, continuous operation, unrestricted
16V4000L64	130 kW/cyl.	3A, continuous operation, unrestricted
12V4000L64FNER	130 kW/cyl.	3A, continuous operation, unrestricted
16V4000L64FNER	130 kW/cyl.	3A, continuous operation, unrestricted

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

© Copyright Rolls-Royce Solutions

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of Rolls-Royce Solutions. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage and/or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

Table of Contents

1	Preface		5.8	Lube oil system	42
1.1	Preface	5	5.9	Cooling system	44
2	Safety		5.10	Engine management and monitoring	47
2.1	Important provisions for all products	6	5.11	Purpose of the units	49
2.2	Intended use of all products	7	5.12	Sensors - Overview	51
2.3	Personnel and organizational requirements	8	6	Technical Data	
2.4	Safety regulations for commissioning and operation	9	6.1	Engine specifications	55
2.5	Safety regulations for commissioning and operation, specific information for gaseous fuel supply or and dual fuel applications	11	6.2	Firing order	56
2.6	Safety regulations for assembly, maintenance, and repair work	13	6.3	Engine - Main dimensions	57
2.7	Safety regulations for assembly, maintenance and servicing work, specific information for gaseous fuel and dual fuel supply or applications	17	7	Operation	
2.8	Fire and environmental protection, indirect materials	18	7.1	Runtimes at partial load	58
2.9	Fire prevention and environmental protection, fluids and lubricants, auxiliary materials, specific information for gaseous fuel supply in gaseous-fuel or dual-fuel applications	21	7.2	Preparations for putting into operation after extended out-of-service periods (>3 months)	59
2.10	Standards for warning messages in the text and highlighted information	22	7.3	Putting the engine into operation after scheduled out-of-service-period	60
3	Transport		7.4	Control, starting and stopping sequences	61
3.1	Transport	23	7.5	Starting the engine	63
3.2	Lifting specifications	24	7.6	Re-starting the engine following an automatic safety shutdown	64
3.3	Crankshaft transport locking device	25	7.7	Operational monitoring	65
4	General Information		7.8	Checking emission values	66
4.1	Engine side and cylinder designations	28	7.9	Engine - Shutdown	67
5	Functional Description		7.10	Engine - Emergency shutdown	68
5.1	Product summary	29	7.11	After stopping the engine - Engine remains ready for operation	70
5.2	Crankcase with oil pan	32	7.12	After shutdown - Putting engine out of operation	71
5.3	Gear train	33	8	Troubleshooting	
5.4	Crank drive	34	8.1	Fault messages - Key	72
5.5	Cylinder head with externally mounted components	36	8.2	Fault messages in engine-generator set controller log	74
5.6	Valve gear	38	8.3	Engine Control Unit - Fault messages	140
5.7	Mixture formation, turbocharging and exhaust system	40	9	Maintenance	
			9.1	50-hours check	182
			9.2	Maintenance task reference table [QL1]	183
			9.3	Maintenance task reference table [QL1]	184
			10	Task Description	
			10.1	Engine	186
			10.1.1	Engine - Barraging manually	186
			10.1.2	Operating room - Check for smell of gas	187
			10.1.3	Engine - Test run	188

10.1.4	Operating data check after 3000 hours	189	10.10.1	Engine oil level - Check	238
10.2	Crankcase Breather	191	10.10.2	Engine oil - Change	239
10.2.1	Oil separator - Filter replacement	191	10.10.3	Engine oil - Sample extraction and analysis	240
10.2.2	Crankcase breather - Pressure regulator check and adjustment	193	10.11	Oil Filtration / Cooling	241
10.3	Ignition System	194	10.11.1	Engine oil filter - Replacement	241
10.3.1	Spark plug - Replacement	194	10.12	Coolant Circuit, General, High-Temperature Circuit	242
10.3.2	Spark plug - Removal	195	10.12.1	Engine coolant - Level check	242
10.3.3	Spark plug - Installation	196	10.12.2	Engine coolant - Change	243
10.3.4	Ignition coil - Rubber cap replacement	198	10.12.3	Engine coolant - Draining	244
10.3.5	Spark plug connector - Replacement	200	10.12.4	Engine coolant - Filling	246
10.3.6	Spark plug connector - Removal	201	10.12.5	Coolant vent line (engine coolant circuit) - Replacement	247
10.3.7	Spark plug connector - Cleaning and check	202	10.12.6	Coolant pipework - O-ring replacement	248
10.3.8	Spark plug connector - Installation	203	10.12.7	Rubber bellows for HT circuit - Replacement	249
10.3.9	Ignition system - Ignition timing check	205	10.13	Low-Temperature Circuit	250
10.4	Valve Drive	206	10.13.1	Mixture coolant - Level check	250
10.4.1	Valve protrusion - Measurement	206	10.13.2	Mixture coolant - Change	251
10.4.2	Valve gear - Lubrication	209	10.13.3	Mixture coolant - Draining	252
10.4.3	Valve clearance - Check and adjustment	210	10.13.4	Mixture coolant - Filling	253
10.4.4	Cylinder head cover - Removal	212	10.14	Engine Mounting / Support	254
10.4.5	Cylinder head cover - Installation	213	10.14.1	Engine mounting - Checking securing screws for firm seating	254
10.5	Gas System	214	10.14.2	Engine mounting - Checking condition of resilient mounts	255
10.5.1	Gas supply - Checking gas lines for leaks	214	10.15	Wiring (General) for Engine/Gearbox/Unit	256
10.6	Exhaust Turbocharger	215	10.15.1	Engine cabling - Check	256
10.6.1	Exhaust turbocharger - Cleaning	215	10.15.2	NOx sensor - Replacement	257
10.7	Air Filter	227	11	Appendix A	
10.7.1	Air filter - Check	227	11.1	Abbreviations	259
10.7.2	Air filter - Replacement	228	11.2	Contact person/Service partner	261
10.7.3	Air filter - Removal and installation	229	12	Appendix B	
10.7.4	Prefilter - Replacement	230	12.1	Standard Tools and Special Tools	262
10.8	Exhaust Pipework	231	12.2	Index	268
10.8.1	Exhaust pipe bellows - Check	231			
10.8.2	Exhaust pipe bellows - Replacement	232			
10.9	Starting Equipment	234			
10.9.1	Starter - Check	234			
10.9.2	Starter - Replacement	235			
10.10	Lube Oil System, Lube Oil Circuit	238			

1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at: <http://www.mtu-solutions.com>

2.2 Intended use of all products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

The product must not be operated in explosive atmospheres unless the engine fulfills the conditions for such use and approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

2.4 Safety regulations for commissioning and operation

Safety regulations for commissioning

Install the product correctly and carry out acceptance in accordance with the manufacturer's specifications before commissioning the product. All necessary approvals must be granted by the relevant authorities and all requirements for commissioning must be fulfilled.

Whenever the product is subsequently put into operation ensure that:

- All personnel is clear of the danger zone surrounding moving parts of the machine.
Electrically-actuated linkages may be set in motion when the Engine Control Unit (governor) is switched on.
- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- All protective devices are in place.
- All lines bearing media are ready for operation.
- The walkways in the operating room are unobstructed and safe.
- No persons wearing pacemakers or any other technical body aids are present.
- Adequate ventilation of the operating room is ensured for any operating status.
- In the first few hours of operation, the product emits gases as a result of smoldering e.g. lacquers or oil. These gases may be hazardous to health. Always wear respiratory protection in the operating room during this period.
- The exhaust system is leak-tight and the gases are vented to atmosphere.
- The product is free of any damage, this applies in particular to lines and cabling.
- All components are properly grounded. Ground separately by means of a grounding stake as necessary.
- The power supply (mains plug) is always freely accessible to facilitate disconnecting the device or system from the power supply at any time.
- Battery terminals, generator terminals or cables are protected against accidental contact.
- All connections have been correctly allocated e.g. +/- polarity, fuel line/reducing agent line, supply/return.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems work properly.

Smoking is prohibited in the area of the product.

Safety regulations during operation

The operator must be familiar with the control and display elements.

The operator must be familiar with the consequences of any operations performed.

During operation, the display instruments and monitoring units must be observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

Practice emergency procedures, especially emergency stopping, at regular intervals.

Take the following steps if any system malfunctions are detected or signaled by the system:

- Inform supervisor(s) in charge.
- Analyze the message.
- Respond by taking any necessary emergency action, e.g. emergency stop.

After a safety shutdown, the engine may only be started after the cause of the shutdown has been eliminated.

Contact Service if the root cause of the malfunction cannot be clearly identified.

Operation

Do not remain in the operating room when the product is running unless absolutely necessary. Keep your stay as short as possible.

Keep a safe distance away from the product if possible. Do not touch the product unless expressly instructed to do so following a written procedure.

Do not inhale the exhaust gases of the product.

The following requirements must be fulfilled before the product is started:

- Wear hearing protection.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binding agent.

Operation of electrical equipment

When electrical equipment is in operation, certain components of these appliances are electrically live.

Follow the applicable operating and safety instructions when operating the devices and heed warnings at all times.